

Assembling the Southeastern Pennsylvania Transportation Authority Asset Inventory



Southeastern Pennsylvania Transportation Authority

Jeffrey D. Knueppel, PE
Assistant General Manager and Chief Engineer

Laura J. Zale
Senior Asset Management Analyst

AECOM

Robert Peskin, Aaron Kozuki, and Adrian Merceron

April 18, 2012

The Southeastern Pennsylvania Transportation Authority



- Sixth largest public transportation agency in the country.
- 1.1 million passenger trips per day.
- Modes Operated:
 - Heavy Rail
 - Commuter (Regional) Rail
 - Light Rail
 - Bus
 - Trackless Trolley
 - Paratransit
- Commuter (Regional Rail) trains also utilize infrastructure owned by Amtrak and CSX.

Operating Territory Infrastructure (All Modes)

- 397.4 Total Miles of Main Line Track
 - 30.8 Total Miles of Elevated Track
 - 58.4 Total Miles of Underground/Tunnel Track
- 90 Interlockings
- 182 Grade Crossings
- 77 Substations/ Switching Stations
- 342 Stations and Bus Terminals
- 23 Vehicle Maintenance Shops



An Intermodal System with almost 3,000 Revenue Vehicles



Infrastructure Origins

- SEPTA was created by the State as many regional transportation providers consolidated between 1964 and 1983.
- Many assets date back to the early 1900's.
- Many assets had fallen into a state of disrepair before SEPTA assumed operations.



Norristown Bridge 6.24 Built in 1882



CHW Bridge 0.35, built in 1915



Callowhill Depot, built in 1913

Working towards SGR is a Core Business Goal.

- Fix-it-First to provide Safe and Reliable Service.
 - Rapid modernization of signal systems ongoing
 - No long term speed restrictions due to track conditions
 - Market-Frankford Line rebuilt end-to-end.
- Maintenance Programs that Prolong Useful Life.



Market Frankford Line West End, circa 1907, rebuilt 2000-2009

Maintaining a Legacy System while Stretching Capital Dollars

- Reverse Engineering of Vehicle Components for VOH Program.
- No “all or nothing” mentality for stations.
- Maintain strong in-house and third party construction programs, including a hybrid approach that utilizes both.



Catenary Replacement on the Commuter Railroad



Allen Lane Station, Reconstructed 2010

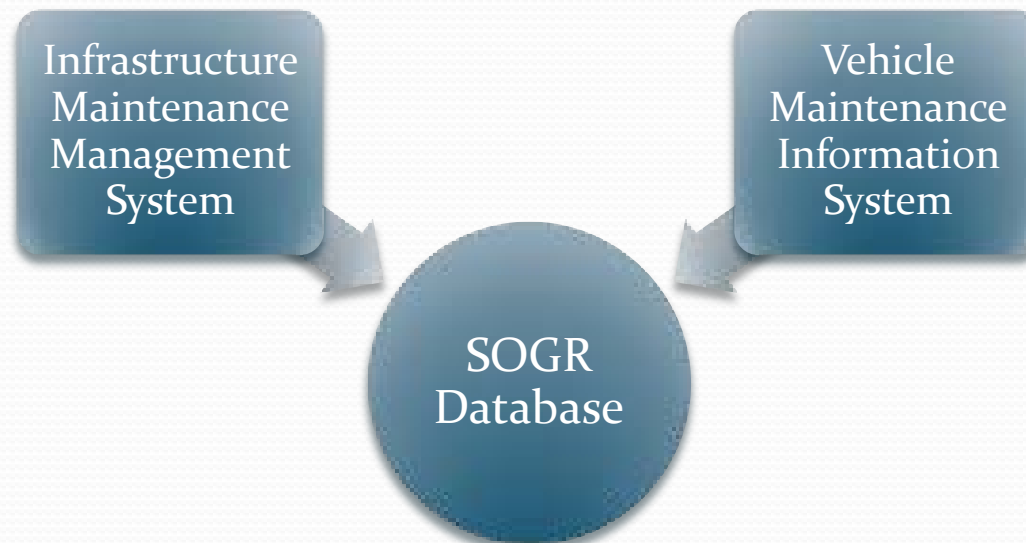
A Need to Prioritize Limited Capital Funds and to Demonstrate our Needs

- SEPTA has lost a large portion of dedicated transportation funding.
- Unfunded mandates (PTC, Radio Narrow Banding, ADA), while well-intended, put additional strain on the capital budget.
- Increased Ridership.
- SEPTA has partnered with Federal, State, and Local officials to demonstrate Capital Needs.
- Transit Asset Management Program started with FTA Grant (November 2010).



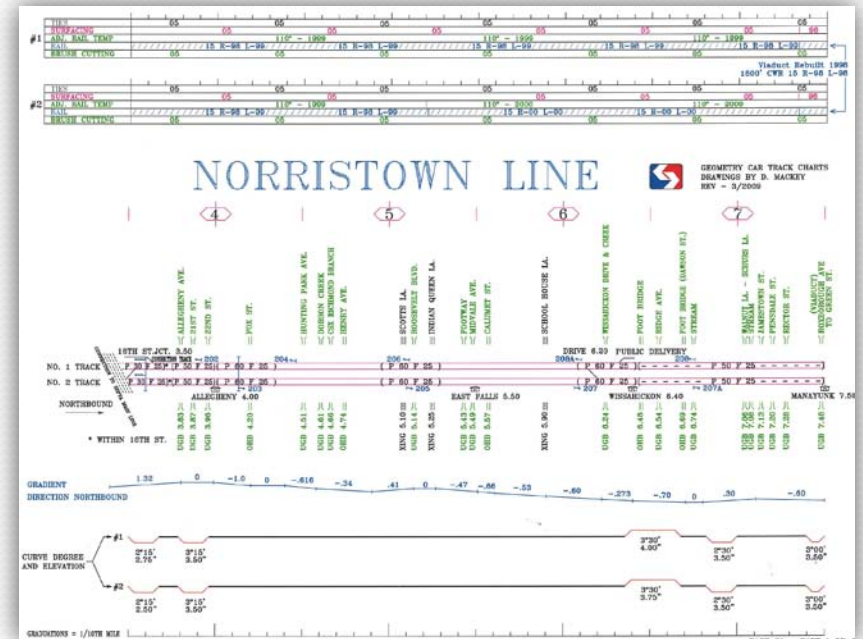
An Inventory-Based Capital Planning Tool: The SGR Database

- SEPTA's SGR database is being developed by AECOM with the goal of:
 - Establishing SEPTA's SGR backlog.
 - Projecting future SGR needs based on asset lifecycles.
 - Prioritizing capital expenditures based on remaining useful life, operational impacts, and cost effectiveness.
- Will utilize data from upgraded vehicle and infrastructure databases but will not be integrated with those systems at this time.



Modeling Assets in the Database

- Age-based rather than condition based.
 - Age and condition data readily available for most asset classes.
 - Simulated condition assessments using TERM decay curves and SEPTA Maintenance experience.



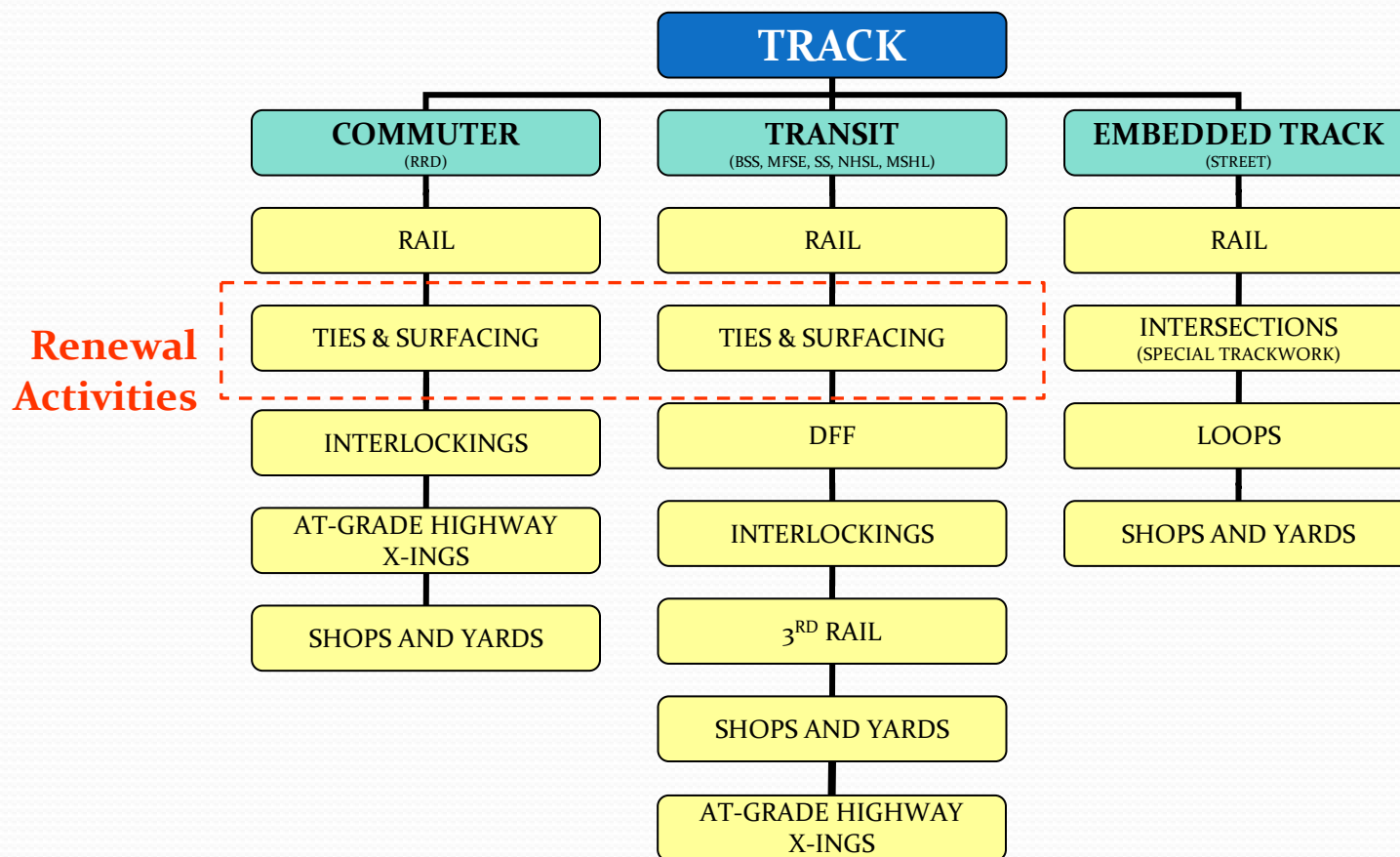
Track maintenance work is recorded in databases and on track charts.

Inventory Assembly Process

- For each identified asset class, AECOM held meetings with various SEPTA staff.
 - Discussed level of information available
 - General condition of assets
- Worked with each department to obtain inventory information.
 - Certain departments had very detailed information that required aggregation
 - Other departments required field reconnaissance
- Asset attributes:
 - Age and Remaining Useful life
 - Renewal activities
 - Cost of renewal activities
 - Cost of Replacement Actions
 - Ridership



Using Maintenance Data to Generate the Inventory



Level of Aggregation

- Philosophy communicated to each department:
 - Group individual assets that have the same following characteristics
 - Service date
 - Useful life
 - Unit replacement cost

Track Asset Record Aggregation Example

Maintenance Management System Output

Asset	Qty	Service Date	Useful Life	Unit Repl. Cost
Curved Rail MP 0-0.2	1,050	1991	15	\$2,200
Curved Rail MP 0.2-0.6	554	1991	15	\$2,200
Straight Rail MP 0-1.0	7,650	1980	40	\$1,000
Straight Rail MP 1.0-2.0	6,340	1980	40	\$1,000
Straight Rail MP 2.0-4.0	15,890	1980	40	\$1,000

SGR Asset Inventory

Asset	Qty	Service Date	Useful Life	Unit Repl. Cost
Curved Rail MP 0-0.6	1,604	1991	15	\$2,200
Straight Rail MP 0-4.0	29,880	1980	40	\$1,000

Capital Planning vs. Maintenance Management Systems

- Identification of key assets by identifying future capital projects.
- Inspection and maintenance cycle information for all assets will be captured in the new Infrastructure and upgraded Vehicle Maintenance Management Systems.

Preliminary Bus Maintenance Facility Inventory

Category	Item	Allegheny	Berridge	Callowhill	Comly	Frankford	Germanatown	Midvale	Southern	Frontier	110 Victory	Total
Building Component	Doors (Overhead Garage)	1		1		23		44			2	72
Building Component	Roofs	1		1	1	1	1	2	1	1	1	10
Building System	(FS) Chemical Sprinkler					3						3
Building System	(FS) Fire Suppression Pumps	1				1		1	1			4
Building System	(FS) Sprinklers	1		1	1	1	1	2	1	1		9
Building System	(HVAC) Air Conditioner	17	1	3	26	25		3	1		3	81
Building System	(HVAC) Heat Recovery Unit	8		9				9				26
Building System	(HVAC) Roof Vents, Fans, Exhaust			1	1	12		44				58
Building System	(HVAC) Unit Heater	1		1		27		1				30
Building System	Backflow	6			5	13	3	2	7	4	7	47
Building System	Boiler	3	2	3	4	4	2	4	4		5	32
Building System	Hot water Heater					1						1
Heavy Maintenance Equipment	Collision Rack		2									2
Heavy Maintenance Equipment	Cranes	4	45	4	3	4	8	3	4	10	6	91
Heavy Maintenance Equipment	Engine Dynamometer		2									2
Heavy Maintenance Equipment	Lathe, Brake and Drum	1					8	1		1		11
Heavy Maintenance Equipment	Lifts, Bus	3	8	8	6	5	9	20	19	4	7	89
Heavy Maintenance Equipment	Lifts, Portable	7	47	1	2	4			1	6	2	70
Heavy Maintenance Equipment	Transmission Dynamometer		1									1
Heavy Maintenance Equipment	Wheel Alignment Machine	1	2		1		1				1	6
Maintenance Equipment	Air Compressor	4	8	4	3	4	4	4	3		7	41
Maintenance Equipment	Air Compressor (Breatheable)										1	1
Maintenance Equipment	Cyclones	3		1	1	2	2		2			11
Maintenance Equipment	Driers	1									1	2
Maintenance Equipment	Emergency Generator	1	1		1	1		2	1			7
Maintenance Equipment	Heater, Paint Booth										1	1
Maintenance Equipment	Paint Booth		2					1	1		2	6
Maintenance Equipment	PLC	5	4	3	6			36	2	1	3	60
Maintenance Equipment	Pressure Vessel	6		1	3	9	1	13	7	3	5	54
Maintenance Equipment	Pressure Washer	3			2	2	1	7	1	1	2	19
Maintenance Equipment	Pumps, Diesel	1		1	1	1	1		1			7
Maintenance Equipment	Pumps, Gasoline	1	1		1	1	1		1		1	7
Maintenance Equipment	Pumps, Water		6	4								10
Maintenance Equipment	Tanks	1	1	1	1	1	1	1	1	1	1	10
Maintenance Equipment	Vacuum, Central System for Busses		1									1
Maintenance Equipment	Washers	2		1	1	3		3	2	1	1	14
Total Items in MP2 Per Shop		86	136	53	74	152	47	207	65	36	62	928

- Building Envelope Assets
 - Roof, Walls, HVAC, Electrical, and Fire Suppression.
- Industrial Equipment Assets
 - Cranes, Lifts, Paint Booth, Vehicle Washer, Transfer Tables

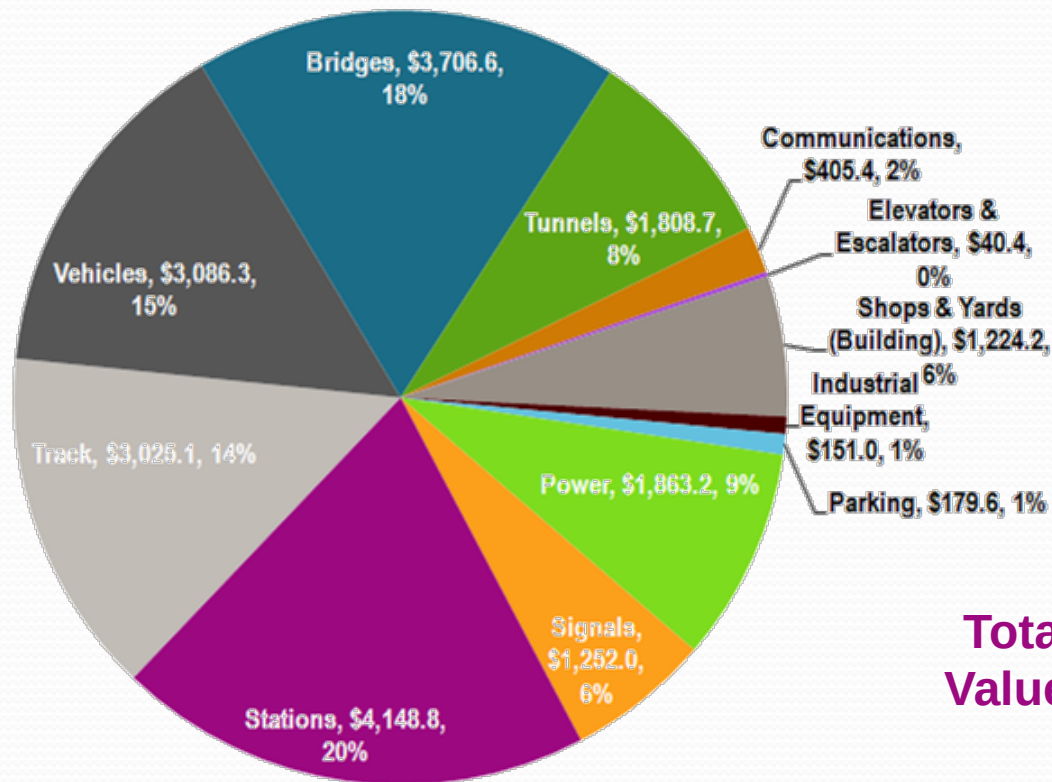
Sample of Asset Inventory

Mode	Line	Asset Type	Asset Group	Asset
Regional Rail	Airport Line	Bridges	Through Girder	Concrete (Deck)
Regional Rail	Airport Line	Bridges	Prestressed Concrete	Concrete (Deck)
Regional Rail	Airport Line	Bridges	Deck Girder	Concrete (Deck)
Regional Rail	Airport Line	Tunnels	Vehicle	Prestressed Concrete
Systemwide	Systemwide	Communications	Equipment	Base Radios
Systemwide	Systemwide	Communications	Cable	Fiber Plant
Subway	Market-Frankford Line	Elevators & Escalators	Frankford Transportation Center	Hydraulic Elevator
Subway	Market-Frankford Line	Elevators & Escalators	Margaret Orthodox	Escalator
Trolley	10, 11, 13, 34, 36	Elevators & Escalators	30th Street	Hydraulic
Subway	Broad Street Line & Spur	Elevators & Escalators	Pattison	Hydraulic
Subway	Broad Street Line & Spur	Elevators & Escalators	AT&T	Escalator
High Speed	Norristown	Elevators & Escalators	Norristown Trans. Center	Hydraulic
Regional Rail	Multiple Regional Rail Lines	Elevators & Escalators	30th Street	Hydraulic
Regional Rail	Airport, Chestnut Hill East, Chestnut Hill West, C	Shops & Yards (Building)	Frazer Shop	Building Envelope
Buses	6,9,27,33,48,54,60,65	Shops & Yards (Building)	Allegheny Bus Garage	Building Envelope
Trolley/Buses	10,15,21,42,30,31,38,40,43,44,46,52,65,MFO	Shops & Yards (Building)	Callowhill Garage	Building Envelope
Subway	Market-Frankford Line	Shops & Yards (Building)	Bridge Street Car House	Building Envelope
Trolley	11, 13, 34, 36	Shops & Yards (Building)	Elmwood Car House	Building Envelope
Buses	C, G, 2, 7, 12, 17, 23, 29, 37, 47, 47M, 57, 64, 6	Shops & Yards (Building)	Southern Garage	Building Envelope
Trolley	10, 11, 13, 34, 36	Shops & Yards (Building)	Woodland Shop	Building Envelope

Number of Asset Records: 6,135

Asset Replacement Value

All Asset Classes (2011\$ Millions)



**Total Replacement
Value = \$20.9 Billion
(2011\$)**

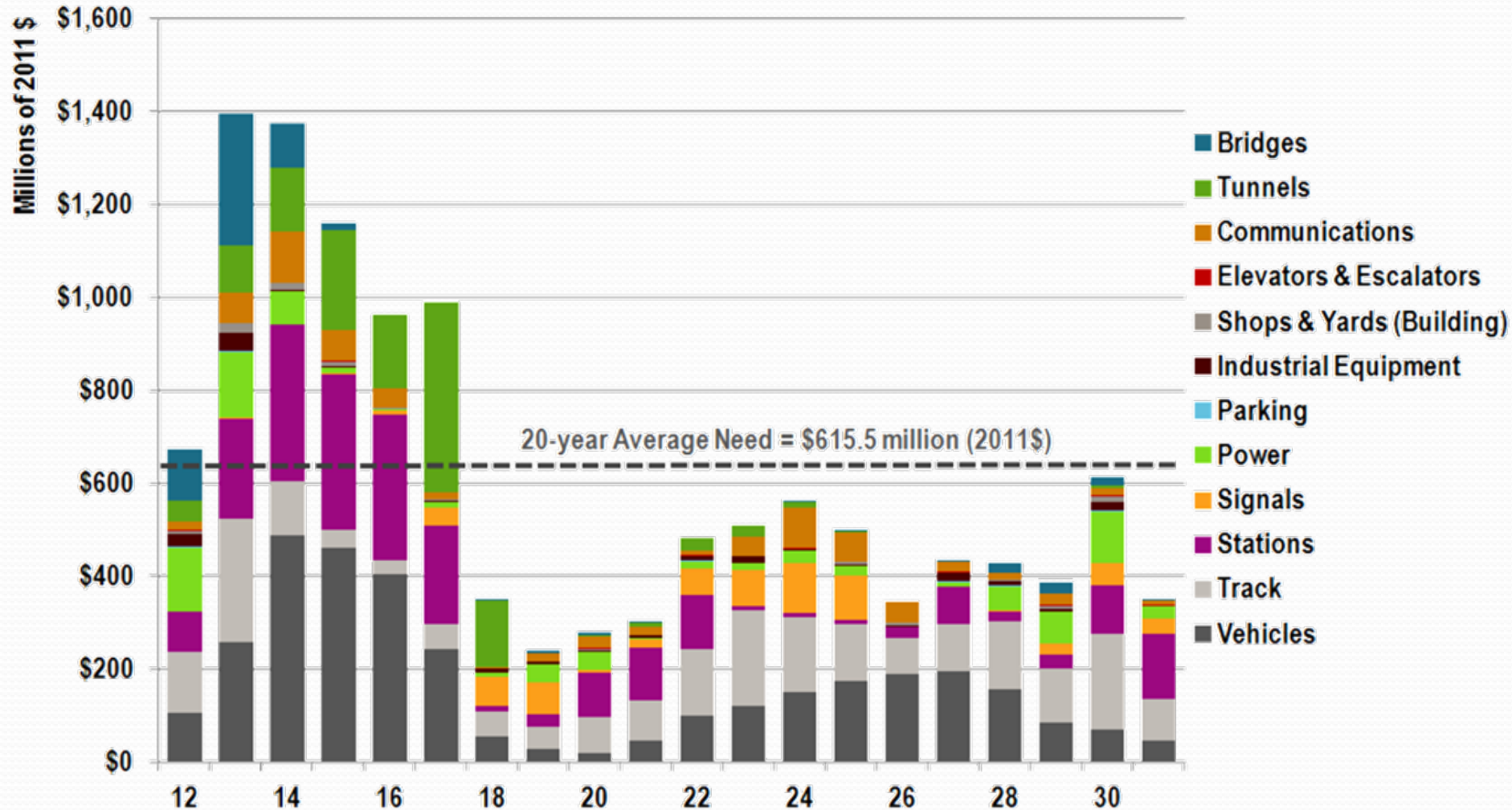
Source: February 2012 Analysis

Preliminary Snapshot of Unconstrained SGR Needs

- Total SGR Backlog Need: \$5.8 B
- Total 20-year SGR Needs: \$12.3 B
- Average Annual SGR Need: \$615.5 M
- Asset Type with Largest 20-year Need: Vehicles (\$3.4 B)
- Number of Replacement Activities over 20-years: 5,857
- Number of Years Needed to Address Backlog: 7 years



Annual SGR Needs by Asset Type (Preliminary)

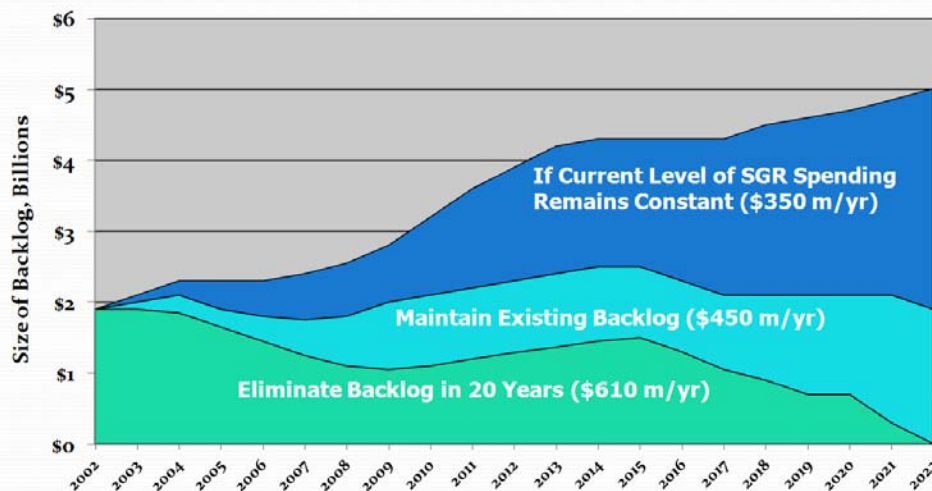


Source: February 2012 Analysis

Next Steps in the SOGR Process

- AECOM is developing the financial modeling portion of the database.
- Determine
 - Funding Level to Maintain Current Backlog.
 - Funding Level Required to Bring the System into a State of Good Repair.
- Illustrate the Cost of Deferred Maintenance.
- Investment Prioritization.

Deferring Maintenance has Financial and Operational Impacts.



Source: MBTA SOGR Database



Catenary failure on Norristown Line, 2011

Lessons Learned from the SOGR Inventory Assembly

- Ask “What is the future project?” when developing a list of capital assets.
- Developing an asset inventory is an iterative and time-consuming task.
- Developing an asset inventory requires buy-in from all departments.
- At SEPTA, this was the first step in assembling data for our future comprehensive Infrastructure Maintenance Management System.
 - Identified deficiencies in data and developed plans for filling them.
- Consultant provided guidance in level of aggregation in addition to data assembly and modeling.



In Conclusion...

- The new SOGR Database will allow SEPTA to
 - Identify measures/funding levels required to bring the System to a State of Good Repair (or at least maintain current condition);
 - Analyze the impacts of various funding and policy scenarios;
 - Provide quantitative analysis for prioritizing/selecting projects for the capital plan; and
 - Articulate the case for additional capital funding.
- It is imperative that SEPTA be able to quantify it's SOGR backlog.
- The new or upgraded transit asset management programs will improve SEPTA's efficiency and effectiveness at maintaining assets.



We are looking forward to giving
an update at the FTA State of Good
Repair Roundtable...
... in Philadelphia!



Jeffrey Knueppel
Laura Zale



Robert Peskin
Aaron Kozuki
Adrian Merceron